

Python

1. What is the output of the following code?

```
x = [1, 2, 3]
y = x.copy()
y.append(4)
print(x)
```

- a) [1, 2, 3, 4]
- b) [1, 2, 3]
- c) [4]
- d) Error

2. Which of the following creates a set comprehension for odd numbers from 1 to 10?

- a) {x for x in range(1, 11) if x % 2 == 0}
- b) {x for x in range(1, 11) if x % 2 != 0}
- c) {x for x in range(1, 10) if x % 2 == 1}
- d) {x for x in range(0, 10) if x % 2 != 0}

3. What does the following code return?

```
def multiply(a, b=3):
    return a * b
print(multiply(4))
```

- a) 12
- b) 4
- c) Error
- d) None

4. Which of the following correctly checks if an element exists in a set?

- a) ``if element in set.keys()``
- b) ``if element in set``
- c) ``if set[element] is not None``
- d) ``if element.exists(set)``

5. What is the output of the following code?

```
lst = [10, 20, 30, 40, 50]
print(lst[-2:])
```

- a) [30, 40]

- b) [40, 50]
- c) [50, 40]
- d) [20, 30]

6. What happens when you run this code?

```
def modify_list(lst):  
    lst = [1, 2, 3]  
    numbers = [4, 5, 6]  
    modify_list(numbers)  
    print(numbers)
```

- a) [1, 2, 3]
- b) [4, 5, 6]
- c) Error
- d) None

7. What will the following code print?

```
try:  
    lst = [1, 2, 3]  
    print(lst[5])  
except IndexError:  
    print("Index out of range")  
except Exception:  
    print("General error")
```

- a) General error
- b) Index out of range
- c) Both messages
- d) Error

8. What is the result of the following dictionary comprehension?

```
{x: x*2 for x in range(1, 5)}
```

- a) {1: 2, 2: 4, 3: 6, 4: 8}
- b) {1: 1, 2: 2, 3: 3, 4: 4}
- c) {2: 2, 4: 4, 6: 6, 8: 8}
- d) {1: 2, 2: 4, 3: 6, 4: 8, 5: 10}

9. Which of the following lambda functions correctly returns the sum of two numbers?

- a) ``lambda x, y: x + y``
- b) ``lambda x, y: x * y``
- c) ``lambda x, y: x - y``
- d) ``lambda: x + y``

10. What is the output of the following code?

```
def outer():  
    x = 10  
    def inner():  
        nonlocal x  
        x += 5  
    inner()  
    return x  
print(outer())
```

- a) 10
- b) 15
- c) Error
- d) 5

Statistics

11. What is the mean of the dataset [2, 4, 6, 8, 10]?

- ☐ A) 5
- ☐ B) 6
- ☐ C) 7
- ☐ D) 8

12. What does the standard deviation measure in a dataset?

- ☐ A) Central tendency
- ☐ B) Spread of data
- ☐ C) Frequency of values
- ☐ D) Correlation between variables

13. What is the median of the dataset [1, 3, 5, 7, 9]?

- ☐ A) 3
- ☐ B) 5
- ☐ C) 7
- ☐ D) 9

14. What is a p-value in hypothesis testing?

- ☐ A) Probability of rejecting the null hypothesis
- ☐ B) Probability of the null hypothesis being true
- ☐ C) Probability of observing the data given the null hypothesis
- ☐ D) Probability of accepting the alternative hypothesis

15. Which measure is affected most by extreme values in a dataset?

- ☐ A) Median
- ☐ B) Mode
- ☐ C) Mean
- ☐ D) Midrange

Machine Learning

16. What is the purpose of a training set in machine learning?

- A) To evaluate the model's performance
- B) To tune hyperparameters
- C) To fit the model
- D) To test the model

17. Which algorithm is commonly used for classification tasks?

- A) Linear Regression
- B) K-Means Clustering
- C) Logistic Regression
- D) Principal Component Analysis

18. What is overfitting in machine learning?

- A) Model performs well on training data but poorly on test data
- B) Model performs poorly on both training and test data
- C) Model is too simple to capture patterns
- D) Model cannot be trained

19. What does the following Scikit-learn code do?

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

- A) Scales the features
- B) Splits data into training and test sets
- C) Fits a model
- D) Evaluates model performance

20. Which metric is used to evaluate a classification model's performance?

- A) Mean Squared Error
- B) R-squared
- C) Accuracy
- D) Root Mean Squared Error

21. What is the purpose of cross-validation in machine learning?

- A) To preprocess data
- B) To reduce model complexity
- C) To assess model performance robustly
- D) To select features

22. What is the output of the following code?

```
from sklearn.preprocessing import StandardScaler
import numpy as np
data = np.array([[1], [2], [3]])
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)
print(scaled_data.mean())
```

- A) 0
- B) 1
- C) 2
- D) Error

23. Which algorithm is an example of unsupervised learning?

- A) Decision Tree
- B) K-Means Clustering

- C) Support Vector Machine
 - D) Linear Regression
24. **What is the role of a loss function in machine learning?**
- A) To split the dataset
 - B) To measure model error
 - C) To select features
 - D) To visualize data
25. **What does the `fit()` method do in Scikit-learn?**
- A) Evaluates the model
 - B) Trains the model
 - C) Predicts new data
 - D) Scales the data
26. **What is the purpose of regularization in machine learning?**
- A) To increase model complexity
 - B) To prevent overfitting
 - C) To speed up training
 - D) To reduce dataset size
27. **Which of the following is a hyperparameter in a Random Forest model?**
- A) Number of trees
 - B) Predicted labels
 - C) Feature weights
 - D) Training data
28. **What does the following code do?**

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

- A) Scales the test data
 - B) Trains and predicts using a logistic regression model
 - C) Performs cross-validation
 - D) Clusters the data
29. **What is the purpose of a confusion matrix?**
- A) To visualize feature importance
 - B) To evaluate classification model performance
 - C) To scale features
 - D) To reduce dimensionality
30. **What is the main advantage of using a decision tree?**
- A) Handles non-linear relationships
 - B) Requires no preprocessing
 - C) Always provides the best accuracy
 - D) Reduces dataset size